FINTECH 545 ASSIGNMENT 2 REPORT

1 Correlations from Mismatched Histories

```
Jointly observed days:
        A    B    C   D  IDX
A     242  242  221  93  242
B     242  242  221  93  242
C     221  221  228  83  228
D      93   93   83  96   96
IDX   242  242  228  96  250
Days all five trade: 81
```

1(a)

The **pairwise method** can return a correlation matrix that is **not PSD**, because each pairwise correlation may be calculated from a different set of jointly observed days. Therefore, the correlations may be mutually inconsistent and can result in a negative eigenvalue. The **complete-row method is guaranteed to be PSD** because all correlations are calculated from the same set of complete observations, so the resulting matrix corresponds to a valid covariance/correlation structure.

1(b)

Eigenvalue tells you how much variance exists in that particular eigenvector direction. Because IDX is nearly a linear combination of the other four series, the true correlation matrix has a smallest eigenvalue close to zero. Therefore, this particular dataset is highly vulnerable to the pairwise method producing a non-PSD correlation matrix.

1(c)

D only started trading partway through the 250-day sample, so it has fewer observations with the other stocks. C trades on a foreign calendar with more holidays, so it also has fewer jointly observed days.
Therefore, **C–D** is likely the least reliable correlation because it has the fewest overlapping observations.

1(d)

```
--- (d) Complete-case correlation ---
          A       B       C       D     IDX
A    1.0000  0.3967  0.2466  0.4136  0.8714
B    0.3967  1.0000  0.3169  0.2361  0.6709
C    0.2466  0.3169  1.0000  0.1852  0.5648
D    0.4136  0.2361  0.1852  1.0000  0.5719
IDX  0.8714  0.6709  0.5648  0.5719  1.0000
eigenvalues: [0.001506 0.548345 0.69332  0.870441 2.886388]
Cholesky: succeeded

--- (d) Pairwise correlation ---
          A       B       C       D     IDX
A    1.0000  0.4686  0.4309  0.4384  0.8696
B    0.4686  1.0000  0.4630  0.2767  0.7329
C    0.4309  0.4630  1.0000  0.1846  0.7136
D    0.4384  0.2767  0.1846  1.0000  0.6057
IDX  0.8696  0.7329  0.7136  0.6057  1.0000
eigenvalues: [-0.011072  0.471422  0.528367  0.857381  3.153902]
Cholesky: FAILED (Matrix is not positive definite)
```

1(e)

```
--- (e) Tracking portfolio ---
Full-history std: {'A': np.float64(0.016502), 'B': np.float64(0.011919), 'C': np.float64(0.018324), 'D': np.float64(0.022829), 'IDX': np.float64(0.012281)}
weights (A,B,C,D,IDX): [-0.4 -0.3 -0.2 -0.1  1. ]
Variance with pairwise Sigma: -1.799370e-06   <-- NEGATIVE: impossible for a true covariance
Reference - realised var(IDX - basket) on complete rows: 3.222052e-07
```

1(f)

```
--- (f) Repaired matrices ---

Rebonato-Jaeckel:
          A       B       C       D     IDX
A    1.0000  0.4691  0.4315  0.4389  0.8620
B    0.4691  1.0000  0.4634  0.2771  0.7278
C    0.4315  0.4634  1.0000  0.1851  0.7082
D    0.4389  0.2771  0.1851  1.0000  0.6014
IDX  0.8620  0.7278  0.7082  0.6014  1.0000

Higham:
          A       B       C       D     IDX
A    1.0000  0.4706  0.4332  0.4402  0.8625
B    0.4706  1.0000  0.4644  0.2777  0.7287
C    0.4332  0.4644  1.0000  0.1858  0.7088
D    0.4402  0.2777  0.1858  1.0000  0.6020
IDX  0.8625  0.7287  0.7088  0.6020  1.0000

Higham converged in 34 iterations

Summary:
                  min_eigenvalue  frobenius_dist  tracking_var  tracking_std
method
Rebonato-Jaeckel   -1.861759e-16    1.618210e-02  6.807298e-07  8.250635e-04
Higham             -4.260125e-13    1.552000e-02  6.787649e-07  8.238719e-04
```

1(g)

A matrix Sigma is PSD if x^T Sigma x >= 0 for every vector x. In other words, every portfolio has non-negative variance. Part (e) found an x that breaks this, so the pairwise Sigma is not the covariance matrix of any random vector. The weights were not random. The cosine between w and the eigenvector of Sigma's negative eigenvalue is 0.999, so the tracking portfolio is the negative-eigenvalue direction. That is the near-singularity from (b). The true variance in that direction is tiny and positive: realised var(IDX - basket) is 3.2 x 10^-7 on the complete days. Inconsistencies of a few hundredths among the IDX correlations were enough to push it negative. In practice, any calculation that uses this covariance and leans on the hedge direction will break: a minimum-variance hedge, a Cholesky-based simulation, or the risk of a hedged book.

1(h)

```
(h) Higham change by entry:
           A       B       C       D     IDX
A     0.0000  0.0020  0.0023  0.0018 -0.0070
B     0.0020  0.0000  0.0014  0.0011 -0.0042
C     0.0023  0.0014  0.0000  0.0012 -0.0048
D     0.0018  0.0011  0.0012  0.0000 -0.0037
IDX -0.0070 -0.0042 -0.0048 -0.0037  0.0000
```

This does not match my answer to (c). The entries I called least reliable (C–D) moved the least. The entries backed by the most data, the IDX row with 228–242 days, moved the most.

Higham's method does not know which correlations are more reliable. It adjusts the correlation matrix to make it positive semidefinite (PSD) using the smallest overall mathematical change.

Higham's repair changes the matrix roughly in proportion to vi vj, where v is this eigenvector.

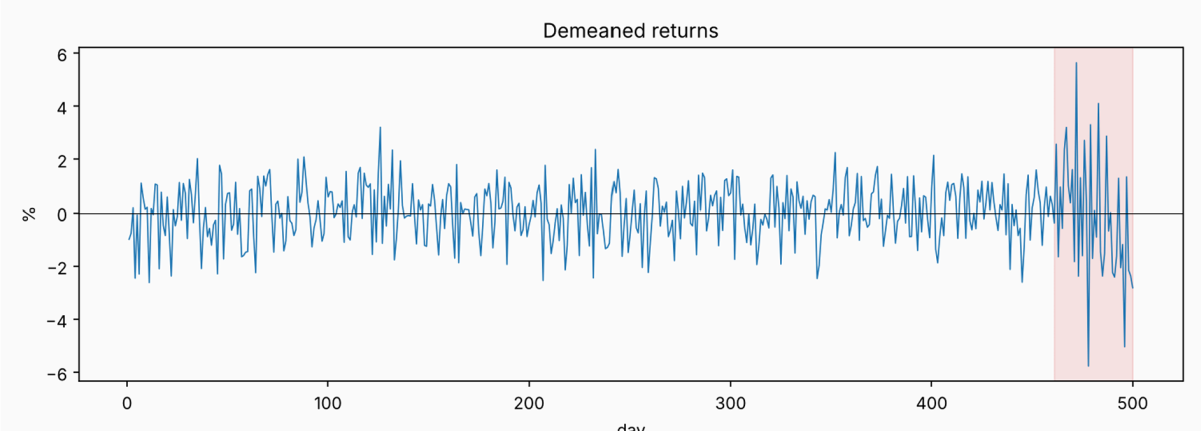
- For stock–index pairs, the eigenvector components have opposite signs, so their correlations decrease.
- For stock–stock pairs, the components have the same sign, so their correlations increase.

1(i)

||Higham - pairwise|| = 0.0155
||complete - pairwise|| = 0.4239
tracking var with complete-case corr: 1.0975452996002874e-06
eigenvector of the negative eigenvalue (A,B,C,D,IDX): [-0.3943 -0.2336 -0.2669 -0.2086 0.8217]

The estimator choice moves the matrix about 27 times further than the repair. For this data, the choice of estimator matters far more than the choice of repair. The complete-case matrix discards 69% of the days, while the pairwise matrix mixes different periods. That trade-off is the decision that drives the answer. The repair is a small final correction to whichever matrix we pick.

2 A Volatility Estimate After the Regime Changed



2(a)

1<5<2<3<4

1: Equal-weighted volatility is the most smoothed, so it should give the lowest volatility estimate if recent volatility is higher.
5: Historical simulation directly uses the actual 5% worst returns, including unusually large losses.
2: lambda = 0.97 puts more weight on recent observations, so it reacts more to recent volatility.
3: lambda = 0.94 puts even more weight on recent observations, so it should produce a higher volatility and therefore higher VaR.
4: Student's t has fatter tails than the Normal distribution, so it generally assigns more probability to extreme losses and gives the largest 5% VaR.

2(b)

A likelihood-based single-break search (rl.variance_breakpoint) puts the change at return #462. That leaves the last 39 days in the new regime, which matches what the plot suggests by eye. The likelihood-ratio statistic against a single variance is 89.8

{'lambda': 0.94, 'n_eff': 32.333333333330934, 'half_life': 11.202305583621158, 'weight_last_39': 0.9104663498474529}
{'lambda': 0.97, 'n_eff': 65.66663473349942, 'half_life': 22.75657306277341, 'weight_last_39': 0.6951421407745042}

2(c)

Moments(n=500, mean=-2.220446049250313e-19
variance=0.00014189813094424568
std=0.011912100190321004
skew=-0.15605505650651794
excess_kurtosis=2.1659417931476597

Under a single normal regime, the excess kurtosis would be zero with a standard error of about $\sqrt{(24/n)} = 0.22$. A value of 2.17 is about 10 standard errors away, so a single normal regime is very unlikely to produce these moments.

2(d)

5% VaR on $1M
1 Normal equal weight: 19593.661202658834
2 Normal EW 0.97 : 34619.07825547442
3 Normal EW 0.94 : 38707.38141459739
4 Student t : 18872.45257735622
5 Historical : 20514.49826306685

Student-t: mu=0.000181623, sigma=0.01029, nu=8.26991 |
logL=1516.6916, AIC=-3027.3831, AICc=-3027.3347, BIC=-3014.7393

2(e)

The result (t < Normal-EQ < Historical < EW 0.97 < EW 0.94) doesn't exactly match my prediction. The t is worth a closer look. It is the smallest even though the data are fat-tailed. A t has more mass in the far tail and less in the shoulders, so at the 5% level its quantile is no further out than the normal quantile for the same data, and usually a little closer to the centre.

2(f)

- **Calm regime:** returns 1–461, which is **461 days**
- **Stress regime:** returns 462–500, the last **39 days**

calm sd: 0.010168448789784545
stress sd: 0.024605580541659195
mixture excess kurtosis: 2.6757534263543405

That is the same order as the 2.17 we measured. The kurtosis in (c) comes from mixing the two regimes, not from heavy tails within either one. So the fitted t describes the unconditional distribution: a random day drawn from the whole history, with a 7.8% chance of being a stress day.
Each regime alone has excess kurtosis about 0 (−0.20 and −0.05), so the fat tails come from *mixing* two normals with different volatilities, not from either regime being fat-tailed.
So the fitted t describes the unconditional distribution: a random day drawn from the whole history, with a 7.8% chance of being a stress day. It does not know that the stress days are the most recent ones, so it does not know today is very likely a stress day. The VaR for "today is in the stress regime" is $40,473, and the calm-regime VaR is $16,726. The t, the equal-weight normal and historical simulation all report something close to the calm number. That is the wrong answer for tomorrow.

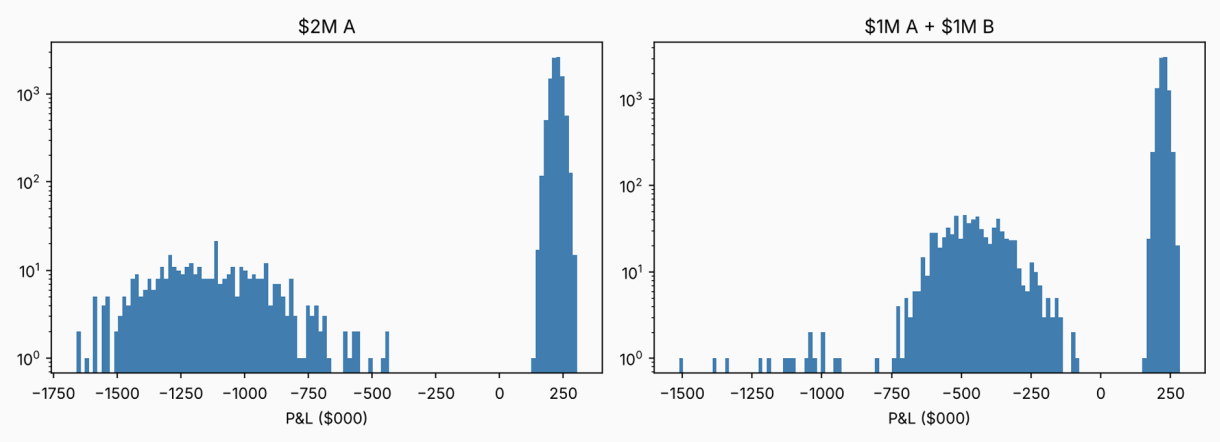
2(g)

lambda 0.97: SE of VaR = 3,021
lambda 0.94: SE of VaR = 4,813
gap between the EW VaRs: 4088.3031591229665
VaR in the stress regime: 40472.57839719471 calm regime: 16725.60987234742

The gap between the two EW VaRs is $4,088 (38707 - 34619) . That is less than one standard error of the λ = 0.94 estimate (0.85 SE) and 0.72 of the combined SE($\sqrt{(4{,}813^2 + 3{,}021^2)}$ = $5,683). The two estimates use the same data and are positively correlated, so the true SE of

the difference is smaller than this. Even so, the gap is not clearly larger than estimation noise. Both estimates say the same thing: risk is now roughly double the full-sample number. Opinion: for a trading limit I would use $\lambda = 0.94$. Limits should react quickly when the regime changes, and a noisier number is acceptable because limits are reviewed every day. For a capital number I would use $\lambda = 0.97$ or longer. Capital should not swing with every two-week move, and the extra effective observations divide the noise by $\sqrt{2}$. I would also check that the capital number is not below a stressed estimate such as the $40k stress-regime VaR.

3 When Diversification Raises VaR



P&L of $2M in A vs $1M in each (log count scale).

3(a)

	A	B
n	10000.000000	10000.000000
mean	0.083709	0.085011
variance	0.018493	0.017451
std	0.135988	0.132102
skew	-4.823246	-4.915516
excess_kurtosis	22.093642	23.086508

This tells you Normal VaR is likely not very reliable for either A or B.

- Both have very negative skewness (A: −4.82, B: −4.92).
  - Returns have a long left tail, meaning extreme losses occur much more often than under a normal distribution.
- Both have very high excess kurtosis (A: 22.09, B: 23.09).
  - There are much fatter tails than the normal distribution, so extreme outcomes are more likely.

- The standard deviations are similar:
  - A: 13.60%, B: 13.21%
- So a Normal VaR calculation based only on volatility would treat them as relatively similar.

3(b)

loss > 20%:
A 406
B 393
at least one 782
both 17

The diversified choice should have **smaller VaR and smaller ES** than the concentrated choice. The fact that only 17 out of 10,000 scenarios have both bonds losing more than 20% shows that the two extreme losses rarely occur together.

3(c)

```
(c) historical, 5%
                    VaR 5%     ES 5%
$1M A            -84570.0  443628.0
$1M B            -86063.0  419587.0
$2M A           -169140.0  887257.0
$1M A + $1M B    414604.0  541997.0
```

3(d)

```
(d) normal, 5%
                Normal VaR 5%  Normal ES 5%
$1M A               139971.0      196794.0
$1M B               132277.0      187477.0
$2M A               279941.0      393589.0
$1M A + $1M B       143466.0      222774.0
```

3(e)

```
(e) historical, 1%
                    VaR 1%      ES 1%
$1M A             649701.0   709548.0
$1M B             640393.0   700805.0
$2M A            1299403.0  1419096.0
$1M A + $1M B     598123.0   716408.0
```

3(f)

Subaddivitiy means $q(P1) + q(P2) \geq q(P1 + P2)$

Measure	ρ(A+B)	ρ(A) + ρ(B)	Result
Historical VaR	$414,604	−$170,633	**Violated**
Historical ES	$541,997	$863,216	Holds

The key is that VaR only looks at the cutoff point, not how severe the losses are beyond that cutoff. Although A and B individually have relatively small 5% VaRs, their extreme losses can occur in different scenarios. When the two positions are combined, the worst 5% of the combined portfolio can contain very large losses from either bond, causing the combined VaR to jump.

ES, however, averages all losses in the worst 5%, so it captures the severity of those extreme losses and properly reflects the diversification benefit.

In 3(b) I predicted the diversified portfolio would have a *smaller* VaR and a smaller ES. The ES prediction was right, but the VaR prediction was wrong. I focused on the fact that both bonds rarely default together (17 scenarios). For VaR, though, the relevant count is how often *at least one* bond defaults (782, or 7.82%), compared with α = 5%. Since 7.82% > 5% while each bond alone is below 5%, combining them moves the 5% quantile into default territory.

3(g)

It favors diversification. However, it is right for the wrong reason. Normal VaR assumes returns are normally distributed. Here, A and B have strong negative skewness and very high excess kurtosis, so the normal assumption misses the extreme downside risk.
The actual historical VaR shows that VaR subadditivity is violated: diversification does not necessarily reduce VaR in this sample. The diversification benefit is better captured by ES, which considers the severity of losses in the tail.

3(h)

At 1%, VaR changes because it reaches the extreme tail where the non-normal behavior of the returns matters most. At 1%, subadditivity holds: the diversified VaR of $598k is below the $1,290k sum. The failure shows up at 5%. That happens because each bond defaults about 4% of the time, *below* α, while at least one of the two defaults 7.8% of the time, *above* α. VaR breaks when α sits between those two numbers.

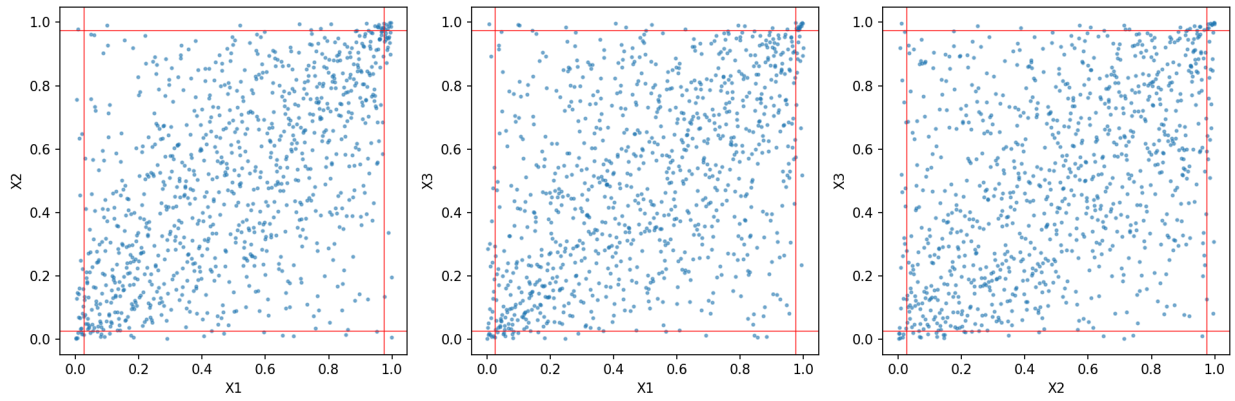
4 Gaussian or t Copula

4(a)

```
(a)
                        X1           X2           X3
n              1000.000000  1000.000000  1000.000000
mean              0.000257    -0.000252     0.000890
variance          0.000141     0.000261     0.000100
std               0.011894     0.016168     0.010019
skew             -0.034226    -0.615040     0.074066
excess_kurtosis  -0.117133     7.124957     2.874089
X1: drop row 951 (-0.0400) -> skew -0.005, excess kurt -0.188
X2: drop row 951 (-0.1429) -> skew 0.056, excess kurt 1.725
X3: drop row 951 (-0.0612) -> skew 0.313, excess kurt 1.731
```

From the moments alone, X1 looks normal and X2 and X3 look fat-tailed, so they need a t. X2's kurtosis of 7.12 and its skew of −0.62 look out of place. Both come from one day, row 951, a −14.3% return that is 8.8 standard deviations from the mean. Without that day, X2's excess kurtosis is 1.73 and its skew is +0.06. The same day is X3's largest move (−6.1%, 6.2 sd). Without it, X3's kurtosis falls from 2.87 to 1.73. That day was also X1's largest move (−4.0%, 3.4 sd), so it was a joint crash.

4(b)



```
     pair  lower  upper
0  X1-X2      6      8
1  X1-X3      7     12
2  X2-X3      6      9
expected if independent: 0.6250000000000001
```

Under independence we would expect 1000×0.025^2 = 0.625 days per pair and side. The series are correlated (Kendall τ 0.31–0.40), so some excess is expected under any copula. The plots show the points crowding into the two diagonal corners more than in the middle of the cloud, and the counts are 10–20 times the independence level. I predicted that the t copula wins.

4(c)

```
        AICc normal       AICc t     chosen          t nu   t sigma
series
X1     -6022.536513 -6020.524453     Normal  1.275916e+13  0.011888
X2     -5408.494036 -5518.184759  Student-t  5.271476e+00  0.012544
X3     -6365.691239 -6428.015517  Student-t  6.094881e+00  0.008181
```

4(d)

```
(d) R =
 [[1.     0.5833 0.5029]
 [0.5833 1.     0.4644]
 [0.5029 0.4644 1.    ]]
```

Copula	**Log-likelihood**	**ν**	**k**	**AICc**	**BIC**
Gaussian	350.44	n/a	3	−694.86	−680.16
t	**452.00**	**3.66**	4	**−895.96**	**−876.37**

4(e)

	VaR 5%	ES 5%	VaR 1%	ES 1%
Gaussian	49410.0	66383.0	76234.0	93827.0
t	48777.0	68281.0	78840.0	102116.0
Historical	47541.0	65699.0	67264.0	102424.0

4(f)

(f) Gaussian

	pair	lower	upper
0	X1-X2	6.39	6.38
1	X1-X3	4.61	4.68
2	X2-X3	4.55	4.49

(f) t

	pair	lower	upper
0	X1-X2	9.17	9.88
1	X1-X3	8.84	8.25
2	X2-X3	7.83	7.40

4(g)

The t copula wins. For Log-likelihood it is higher than gaussian by 101.6. Its AICc and BIC is lower. For tail counts, the Gaussian implies 31 joint-tail days in total against 48 in the data, falling short most in the upper tail (X1–X3: 4.7 vs 12). The t implies 51, close to the data overall. The t slightly overstates the lower joint tails (7.8–9.2 vs 6–7) and understates the X1–X3 upper tail. The data's joint tails are asymmetric, with more joint good days than joint bad days, and neither elliptical copula can capture that. Each count is small (Poisson sd ≈ 2.5–3.5), so differences of 2–3 days per pair are within noise. The total across pairs is the more reliable comparison, and there the t is clearly better

4(h)

ES at 1% moves the most. The 5% VaR barely moves. Both copulas share the same margins and the same R, so the body of the portfolio distribution is almost the same under each. A 5% quantile is set by that body: ordinary days on which the three returns are moderately correlated. Tail dependence only affects outcomes in which all three assets are extreme at once. Those scenarios lie beyond the 1% quantile, and ES at 1% averages them. The t copula also puts more mass at the centre and in the corners, and less in between, so its 5% VaR is even slightly lower. Historical ES at 1% is ($102,426) far better than the Gaussian ($93,827).

4(i)

total gain: 101.559
days with NO series in outer 5% : 793 contribution: 50.478
days with >=1 series in outer 5%: 207 contribution: 51.081
biggest single days: [641 480 86] [10.294 10.251 6.097]
without row 951: nu = 3.67 AICc gap = 195.885

I expected the t-copula to perform better because it can capture days when multiple assets crash together. However, about half of its advantage comes from ordinary days, because it models the concentration of observations near the centre of the distribution better than the Gaussian copula.

AICc evaluates how well the model fits the entire distribution, including ordinary days and extreme events in all directions. But for a long portfolio, we're especially concerned with large losses in the lower tail.

Therefore, a t-copula can win the AICc comparison without necessarily estimating downside risk accurately.

4(j)

The most correlated pair is X1 - X2 (ρ = 0.583) . lambda_t(X1,X2) = 0.3219
Given that one asset has a very extreme day, the probability that the other does too stays near one third, however far into the tail we go. Under the Gaussian copula the tail dependence is 0

5 Model Based Simulation and Residual Correlation

5(a)

$$\text{For dollar weights } (w_A, w_B):$$
$$\text{Var}(P) = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\,\text{Cov}(r_A, r_B), \qquad \text{Cov}(r_A, r_B) = \beta_A\beta_B\sigma_M^2 + \text{Cov}(\epsilon_A, \epsilon_B)$$

If the residuals ϵA and ϵB are incorrectly assumed independent, their covariance is set to zero, potentially missing the common industry risk.

P1: Here wA,wB>0. If the residuals are positively correlated because of shared industry risk, ignoring their covariance understates portfolio variance and VaR. So P1 is understated

P2: Here wA>0 and wB<0. The covariance term is negative, so ignoring positive residual covariance overstates portfolio variance and VaR by missing the additional hedging benefit. So P2 is overstated.

5(b)

```
        alpha      beta  resid_sd
A  0.000205  1.035278  0.012888
B  0.000315  0.935417  0.011538
residual correlation: 0.6983
```

5(c)

```
(c)
                              P1       P2
full residual cov        50239.0  15845.0
independent residuals    44567.0  28532.0
```

5(d)

delta normal VaR P1: 50,415
delta normal VaR P2: 15,834

5(e)

$$\mathrm{Var}(P) = \underbrace{(\beta_A \pm \beta_B)^2 \sigma_M^2}_{\text{market}} + \underbrace{\sigma_{\epsilon_A}^2 + \sigma_{\epsilon_B}^2 \pm 2\,\mathrm{Cov}(\epsilon_A, \epsilon_B)}_{\text{residual}}$$

Yes, both moved in the predicted direction. P2 is more exposed to the assumption because the stocks' market betas nearly cancel out: 1.035−0.935=0.10
This makes P2 almost market-neutral. Its risk depends heavily on the residual correlation, so ignoring it has a large impact. The residual correlation matters most when the portfolio relies on one stock hedging the other.

5(f)

The closed form of full residual covariance VaR is *identical* to the delta normal VaR. The simulated values differ by less than 0.4%, which is Monte Carlo noise.

OLS with an intercept forces the residuals to have zero sample covariance with the market. The full factor model includes both the market risk and the residual risk. The covariance between the two stocks is

$$\mathrm{Cov}(r_A, r_B) = \beta_A \beta_B \sigma_M^2 + \mathrm{Cov}(\epsilon_A, \epsilon_B)$$

The first term represents their shared market risk, and the second represents their remaining covariance. Including the full residual covariance means no covariance information is lost.

If the market return and the stock returns are jointly normal, then the stocks' returns conditional on the market are also jointly normal. Their conditional means are linear in the market return, with slopes equal to the OLS betas, and their residual covariance is constant.

So the regression model with full residual covariance is just the joint normal written in two steps (market first, then stocks given market). It adds no information and removes none, so it must give the same VaR as the joint normal estimated directly.

The two only diverge when the model restricts something, as the diagonal shortcut does, or assumes something beyond normality.